

```

81 • SELECT order_date,
82         SUM(total_amount) AS daily_sales
83     FROM orders
84     WHERE total_amount IS NOT NULL
85     GROUP BY order_date
86     ORDER BY order_date;
87
88     SELECT c.city,

```

Result Grid |   Filter Rows: | Export:  | Wrap Cell Content: 

order_date	daily_sales
2024-01-02	1050.00
2024-01-06	125.00
2024-01-22	240.00
2024-01-28	25.00
2024-02-05	120.00
2024-02-06	1600.00
2024-02-07	3200.00
2024-02-08	225.00
2024-02-10	100.00
2024-02-11	1750.00

Result 18

```

88     SELECT c.city,
89             SUM(o.total_amount) AS total_revenue
90     FROM customers c
91     JOIN orders o
92     ON c.customer_id = o.customer_id
93     WHERE c.city IS NOT NULL
94     GROUP BY c.city
95     ORDER BY total_revenue DESC;

```

Result Grid |   Filter Rows: | Export:  | Wrap Cell

	city	total_revenue
►	Chicago	8335.00
	Seattle	4125.00
	Detroit	3510.00
	San Diego	3500.00
	Baltimore	3435.00
	Boise	3385.00
	Milwaukee	3040.00
	Las Vegas	2900.00
	Indianapolis	2810.00
	Honolulu	2730.00

```

97
98 • SELECT c.customer_id,
99         CONCAT(c.first_name, ' ', c.last_name) AS customer_name,
100         COUNT(o.order_id) AS order_count
101 FROM customers c
102 JOIN orders o
103 ON c.customer_id = o.customer_id
104 GROUP BY c.customer_id, c.first_name, c.last_name
105 HAVING COUNT(o.order_id) > 2;

```

Result Grid |  Filter Rows: | Export:  | Wrap Cell Content: 

	customer_id	customer_name	order_count
▶	3	Olivia Brown	3
	12	William Anderson	3
	14	Ethan Taylor	4
	24	Daniel Walker	5
	25	Harper Hall	4
	26	Matthew Allen	4
	31	Chloe Adams	3
	33	Grace Baker	4
	36	Mason Campbell	5
	37	Avery Allen	3

Result 20 ▾

```

107
108 • SELECT city,
109         customer_id,
110         customer_name,
111         total_spend
112 FROM (
113     SELECT
114         c.city,
115         c.customer_id,

```

Result Grid |  Filter Rows: | Export:  | Wrap Cell Content: 

	city	customer_id	customer_name	total_spend
▶	Albany	9	Sophia Garcia	290.00
	Atlanta	16	Jackson White	390.00
	Baltimore	36	Mason Campbell	3435.00
	Boise	7	Isabella Davis	3385.00
	Boston	22	Henry Moore	160.00
	Centerville	2	Emma Jones	300.00
	Charlotte	19	Madison Thompson	1180.00
	Chicago	43	Lily Turner	8335.00
	Columbus	30	David Scott	25.00
	Dallas	48	James Walker	800.00

Result 21 ▾

```

.34 • SELECT
.35     c.customer_id,
.36     CONCAT(c.first_name, ' ', c.last_name) AS customer_name,
.37     c.city,
.38     SUM(o.total_amount) AS total_spend,
.39     COUNT(o.order_id) AS order_count
.40 FROM customers c
.41 JOIN orders o

```

Result Grid |   Filter Rows: | Export:  | Wrap Cell Content: 

	customer_id	customer_name	city	total_spend	order_count
	43	Lily Turner	Chicago	8335.00	5
	46	Michael Cooper	Seattle	4000.00	1
	36	Mason Campbell	Baltimore	3435.00	5
	7	Isabella Davis	Boise	3385.00	2
	49	Elena Gray	Detroit	3290.00	2
	26	Matthew Allen	Milwaukee	3040.00	4
	25	Harper Hall	Honolulu	2730.00	4
	41	Penelope Roberts	Las Vegas	2720.00	3
	50	Henry Phillips	Indianapolis	2450.00	3
	47	Amelia James	San Antonio	2280.00	3

Result 22 x

1. Read sales data → clean nulls → calculate daily sales

```
from pyspark.sql.functions import sum as _sum, col

daily_sales = (orders
               .filter(col("total_amount").isNotNull())
               .groupBy("order_date")
               .agg(_sum("total_amount").alias("daily_sales"))
               .orderBy("order_date"))

daily_sales.show()
```

```
... +-----+-----+
|order_date|daily_sales|
+-----+-----+
|2024-01-02|    1050.0|
|2024-01-06|     125.0|
|2024-01-22|     240.0|
|2024-01-28|      25.0|
|2024-02-05|     120.0|
|2024-02-06|    1600.0|
|2024-02-07|    3200.0|
|2024-02-08|     225.0|
|2024-02-10|     100.0|
|2024-02-11|    1750.0|
|2024-02-23|     135.0|
|2024-02-25|     480.0|
|2024-02-27|    2400.0|
|2024-02-29|     160.0|
|2024-03-02|     240.0|
|2024-03-04|     400.0|
|2024-03-06|     880.0|
|2024-03-07|     160.0|
|2024-03-08|     320.0|
|2024-03-12|     125.0|
+-----+-----+
only showing top 20 rows
```

2. Read customer data -> clean invalid rows -> city-wise revenue

```
city_revenue = (customers.filter(col("city").isNotNull())
    .join(orders, "customer_id")
    .groupBy("city")
    .agg(_sum("total_amount").alias("total_revenue"))
    .orderBy(col("total_revenue").desc()))
city_revenue.show()
```

```
+-----+-----+
|      city|total_revenue|
+-----+-----+
|   Chicago|      8335.0|
|   Seattle|      4125.0|
|   Detroit|      3510.0|
| San Diego|      3500.0|
| Baltimore|      3435.0|
|    Boise|      3385.0|
| Milwaukee|      3040.0|
| Las Vegas|      2900.0|
| Indianapolis|      2810.0|
| Honolulu|      2730.0|
|    Denver|      2400.0|
| San Antonio|      2280.0|
|   Charlotte|      2185.0|
|    El Paso|      2100.0|
|   New York|      1950.0|
|   Memphis|      1705.0|
| Minneapolis|      1635.0|
| Springfield|      1600.0|
|   San Jose|      1450.0|
|   St. Louis|      1445.0|
+-----+-----+
only showing top 20 rows
```

3. Find repeat customers (>2 orders)

```
from pyspark.sql.functions import count, col

repeat_customers = (customers.join(orders, "customer_id")
    .groupBy("customer_id", "first_name", "last_name")
    .agg(count("order_id").alias("order_count"))
    .filter(col("order_count") > 2))
repeat_customers.show()
```

```
... +-----+-----+-----+-----+
|customer_id|first_name|last_name|order_count|
+-----+-----+-----+-----+
|      12|   William|  Anderson|          3|
|      41| Penelope|   Roberts|          3|
|      36|    Mason| Campbell|          5|
|      25|   Harper|    Hall|          4|
|      50|    Henry| Phillips|          3|
|       3|   Olivia|   Brown|          3|
|      43|    Lily|   Turner|          5|
|      37|    Avery|    Allen|          3|
|      14|    Ethan|   Taylor|          4|
|      24|   Daniel|   Walker|          5|
|      47|   Amelia|    James|          3|
|      26| Matthew|    Allen|          4|
|      31|    Chloe|   Adams|          3|
|      33|    Grace|   Baker|          4|
|      39|    Aria|    Davis|          3|
+-----+-----+-----+-----+
```


4. Find highest spending customer in each city

```
from pyspark.sql.window import Window
from pyspark.sql.functions import rank, sum as _sum, col

customer_spend = (customers.join(orders, "customer_id")
    .groupBy("city", "customer_id", "first_name", "last_name")
    .agg(_sum("total_amount").alias("total_spend")))

city_window = Window.partitionBy("city").orderBy(col("total_spend").desc())

top_per_city = (customer_spend
    .withColumn("rnk", rank().over(city_window))
    .filter(col("rnk") == 1)
    .drop("rnk"))
top_per_city.show()
```

```
***
+-----+-----+-----+-----+-----+
| city|customer_id|first_name|last_name|total_spend|
+-----+-----+-----+-----+-----+
| Albany|9|Sophia|Garcia|290.0|
| Atlanta|16|Jackson|White|390.0|
| Baltimore|36|Mason|Campbell|3435.0|
| Boise|7|Isabella|Davis|3385.0|
| Boston|22|Henry|Moore|160.0|
| Centerville|2|Emma|Jones|300.0|
| Charlotte|19|Madison|Thompson|1180.0|
| Chicago|43|Lily|Turner|8335.0|
| Columbus|30|David|Scott|25.0|
| Dallas|48|James|Walker|800.0|
| Denver|40|Benjamin|Evans|1820.0|
| Des Moines|8|James|Martinez|445.0|
| Detroit|49|Elena|Gray|3290.0|
| El Paso|38|Logan|Carter|2100.0|
| Greenville|3|Olivia|Brown|675.0|
| Honolulu|25|Harper|Hall|2730.0|
| Houston|42|Elijah|Young|135.0|
| Indianapolis|50|Henry|Phillips|2450.0|
| Jackson|33|Grace|Baker|220.0|
| Jacksonville|21|Scarlett|Wilson|80.0|
+-----+-----+-----+-----+-----+
only showing top 20 rows
```

5. Build final reporting table with customer, city, total spend, order count

```
from pyspark.sql.functions import sum as _sum, col, count

reporting_table = (customers.join(orders, "customer_id")
    .groupBy("customer_id", "first_name", "last_name", "city")
    .agg(
        _sum("total_amount").alias("total_spend"),
        count("order_id").alias("order_count")
    )
    .orderBy(col("total_spend").desc()))
reporting_table.show()
```

```
*** +-----+-----+-----+-----+-----+-----+
|customer_id|first_name|last_name|city|total_spend|order_count|
+-----+-----+-----+-----+-----+-----+
|43|Lily|Turner|Chicago|8335.0|5|
|46|Michael|Cooper|Seattle|4000.0|1|
|36|Mason|Campbell|Baltimore|3435.0|5|
|7|Isabella|Davis|Boise|3385.0|2|
|49|Elena|Gray|Detroit|3290.0|2|
|26|Matthew|Allen|Milwaukee|3040.0|4|
|25|Harper|Hall|Honolulu|2730.0|4|
|41|Penelope|Roberts|Las Vegas|2720.0|3|
|50|Henry|Phillips|Indianapolis|2450.0|3|
|47|Amelia|James|San Antonio|2280.0|3|
|44|Daniel|Morris|San Diego|2200.0|2|
|38|Logan|Carter|El Paso|2100.0|2|
|45|Charlotte|Wood|New York|1950.0|1|
|40|Benjamin|Evans|Denver|1820.0|2|
|39|Aria|Davis|Memphis|1705.0|3|
|14|Ethan|Taylor|Minneapolis|1635.0|4|
|1|John|Smith|Springfield|1600.0|1|
|31|Chloe|Adams|San Jose|1450.0|3|
|24|Daniel|Walker|St. Louis|1445.0|5|
|17|Charlotte|Harris|San Diego|1300.0|1|
+-----+-----+-----+-----+-----+-----+
only showing top 20 rows
```